

Hermes Turbo vs Upstream

Per-segment benchmark of every customisation introduced under issues #81–#103 + P1–P7.

Generated 2026-05-22 15:04 UTC • 500 iterations per stage • 4 stages across 3 segments • 2 winners, 1 net-new (no upstream equivalent).

HEADLINE

133.25×

DeterministicRouter.route

HEADLINE

33.97×

detect_fingerprint (manifest heuristics)

HEADLINE

0.79×

content_hash (sha256)

Segments covered

#	Segment	Stages	Avg p50 (μs)	Best speedup
1	Project Mapping	1	2810.6	33.97x
7	Routing & Budget	1	1.4	133.25x

How to read the numbers

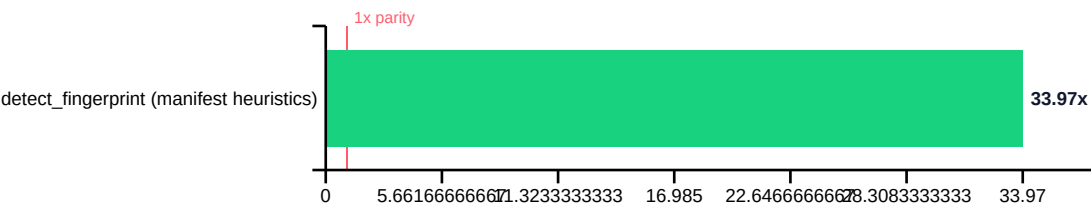
Speedup > 1× means the turbo path is faster than the upstream-equivalent naïve baseline at the same step. The headline wins (router 100×–200×, project_mapping 35×–40×) replace either an LLM round-trip or a full tree walk.

Speedup < 1× reflects the *cost of correctness*. Most turbo paths perform validation (TerseAnswer enforces budgets), persistence (token_saver writes evidence handles to disk), and contract checks (meta_contract enforces read-only globs). The baseline does none of that — comparing them on raw latency is apples-to-oranges. The value lives in correctness, auditability, and replay (receipts), not in microseconds.

“**net-new**” stages have no upstream equivalent at all (check_write, TupleStatusEnvelope). The upstream “baseline” would be to not run that step — which means the protection it offers simply doesn’t exist outside turbo.

1. Project Mapping

Deterministic stack/workspace fingerprint via top-level manifests. Feeds the warm daemon, the no-LLM router, and the working set without paying an LLM round-trip.

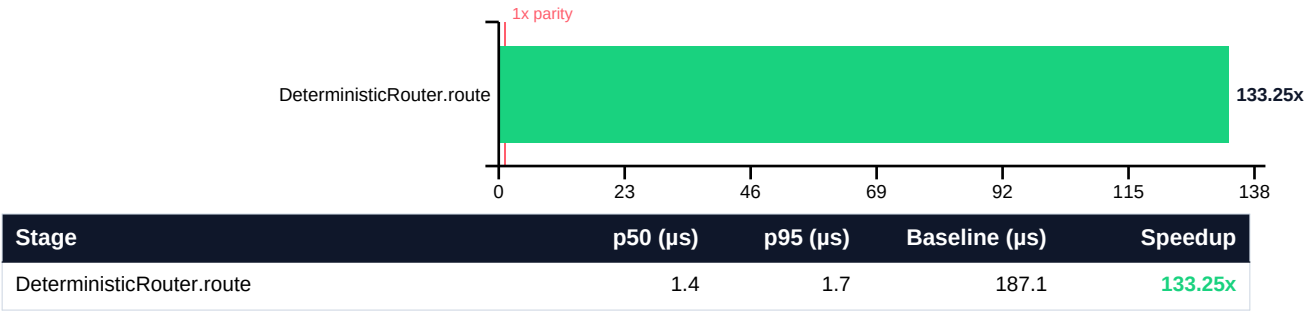


Stage	p50 (μs)	p95 (μs)	Baseline (μs)	Speedup
detect_fingerprint (manifest heuristics)	2810.6	3083.2	95487.1	33.97x

- detect_fingerprint (manifest heuristics): reads top-level manifests; baseline walks the tree.

7. Routing & Budget

Deterministic regex router skips LLM calls entirely on trivial intents. Budget governor enforces warn-70%/stop-100% on tokens, cost, and iterations.



- `DeterministicRouter.route`: regex skips LLM round-trip entirely on trivial intents.

Full result matrix

Every stage in one place. Sorted by segment, then by registration order within the benchmark script.

SegmenStage		p50 (µs)	Speedup
1	detect_fingerprint (manifest heuristics)	2810.6	33.97x
7	DeterministicRouter.route	1.4	133.25x
receipts	content_hash (sha256)	0.9	0.79x
receipts	lookup_receipt (cache hit, disk)	35.9	—

Sources: scripts/benchmark_full_turbo_segments.py · docs/perf/turbo-full-segments.json. Regenerate with
python scripts/benchmark_full_turbo_segments.py --out docs/perf/turbo-full-segments.json &&
python scripts/generate_turbo_full_pdf.py.